

iStudio

Name: Ayush Kumar

Assignment number: 1

Course / Internship name: **Data Analytics Internship**

Task 1:

Excel file showing the formatted dataset, total_amount column, 7 calculated function values, and the 3 charts

Chart 1 (Bar graph)

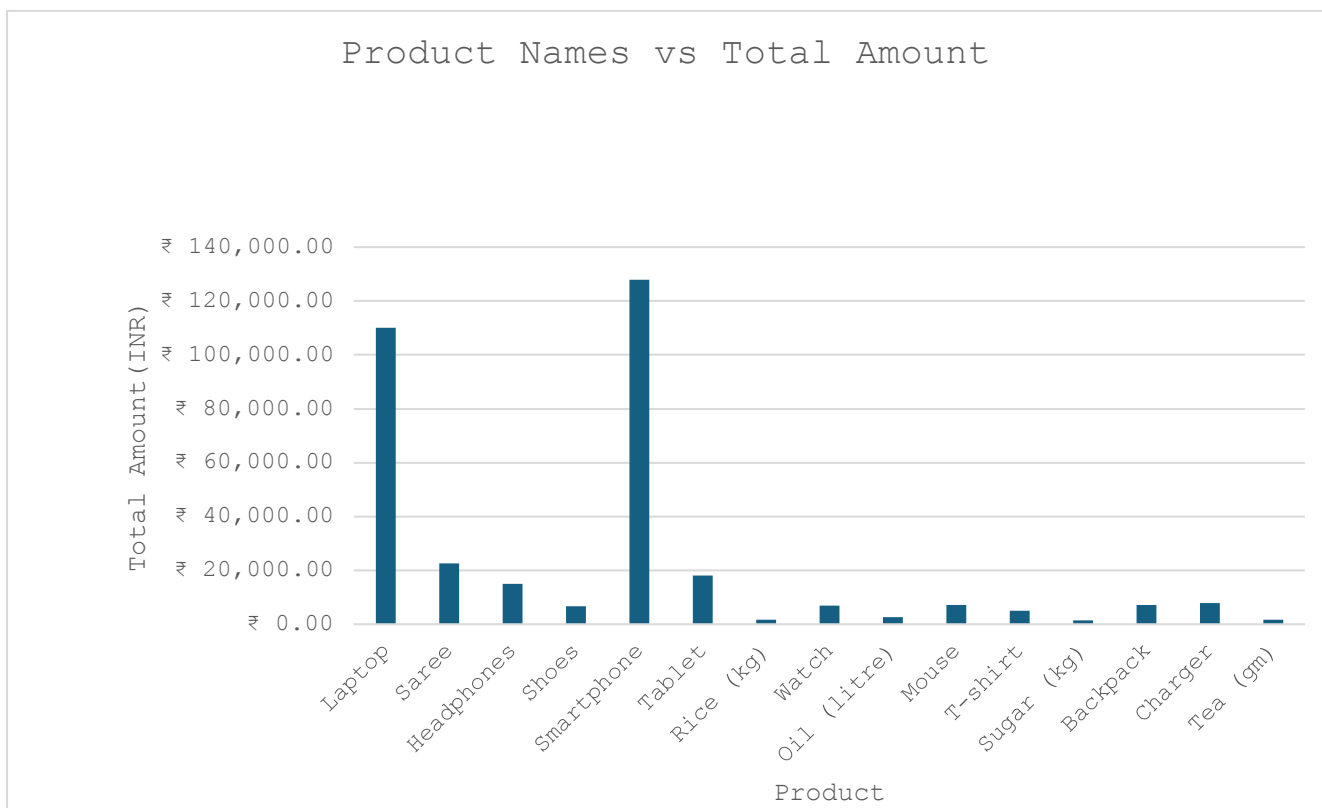


Chart 2 (Pie chart)

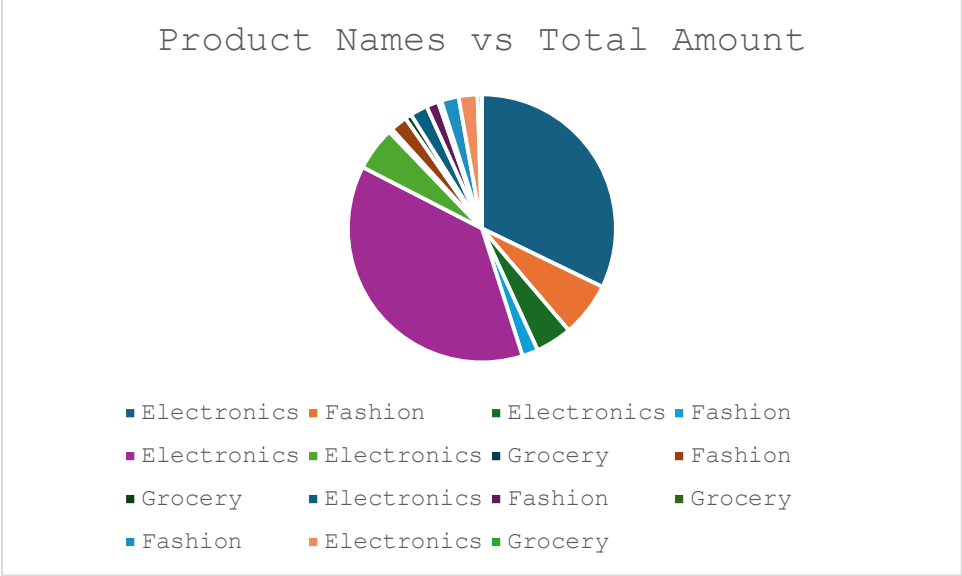
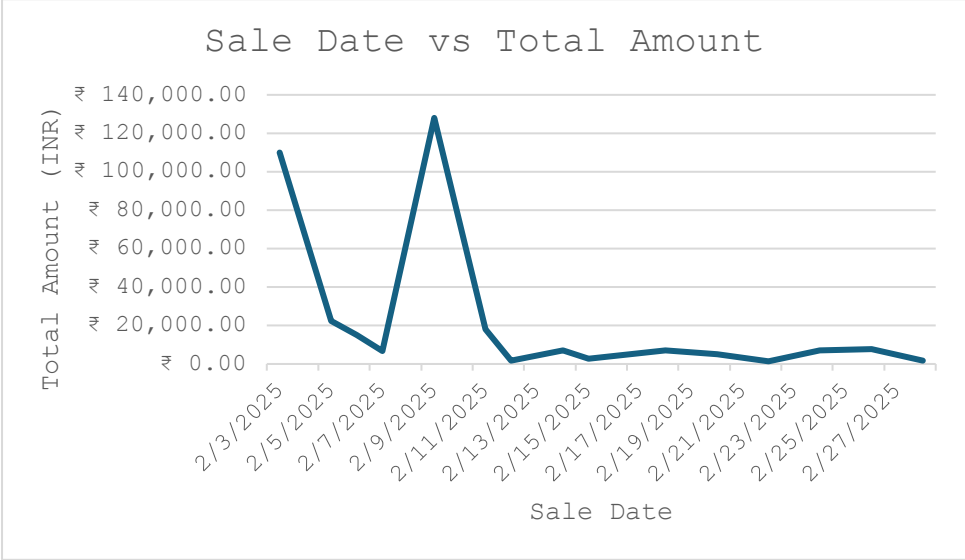


Chart 3 (Line graph)



Formula table

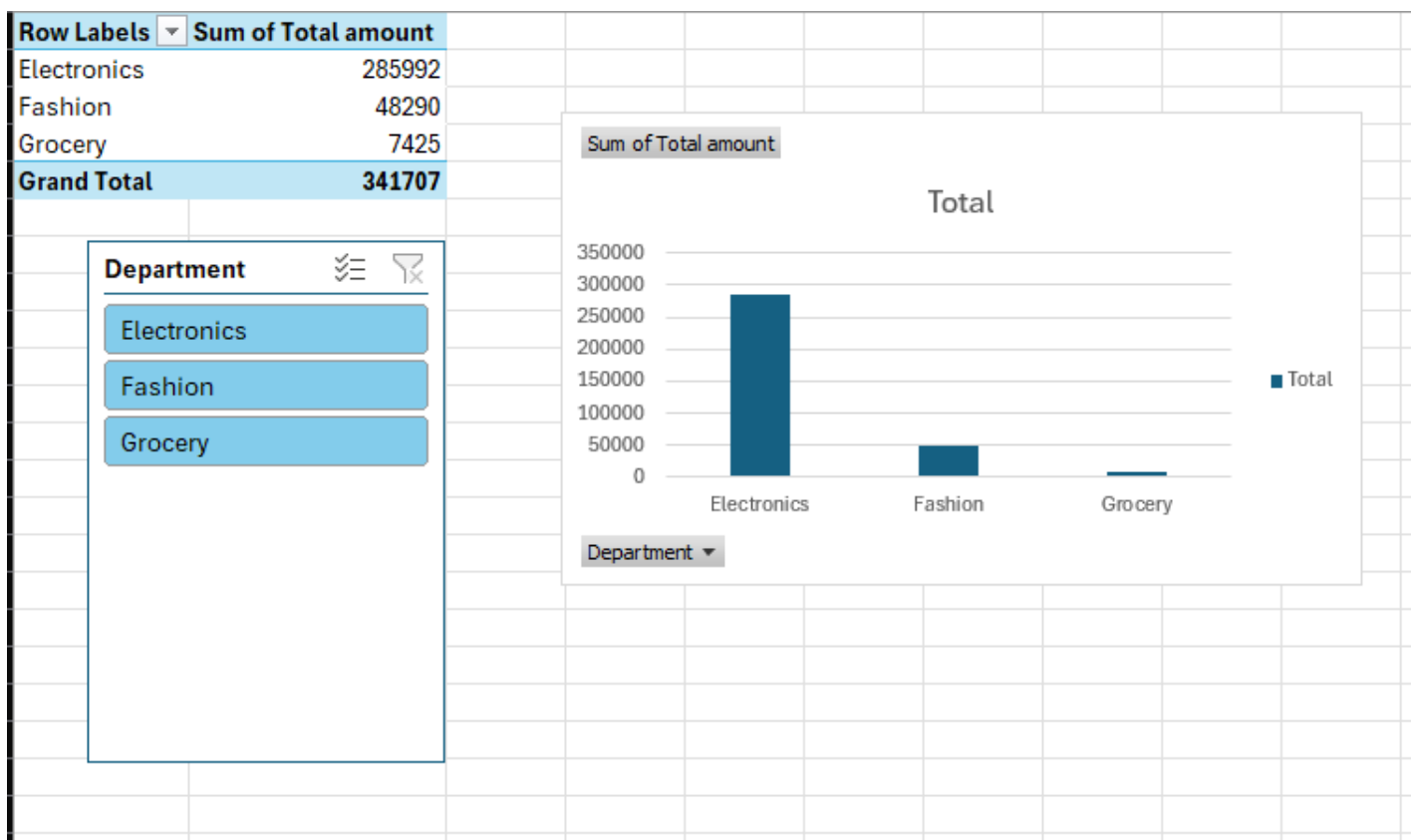
#	Label Text	Excel Formula
1.	Total Revenue:	=SUM(H2:H16)
2.	Average Sale:	=AVERAGE(H2:H16)
3	Maximum Sale:	=MAX(H2:H16)
4.	Minimum Sale:	=MIN(H2:H16)
5.	Fashion Sales Count:	=COUNTIF(C2:C16, "Fashion")

6.	Electronics Revenue:	=SUMIF(C2:C16, "Electronics", H2:H16)
7.	Bonus	=IF(H2>50000, "High", "Normal")

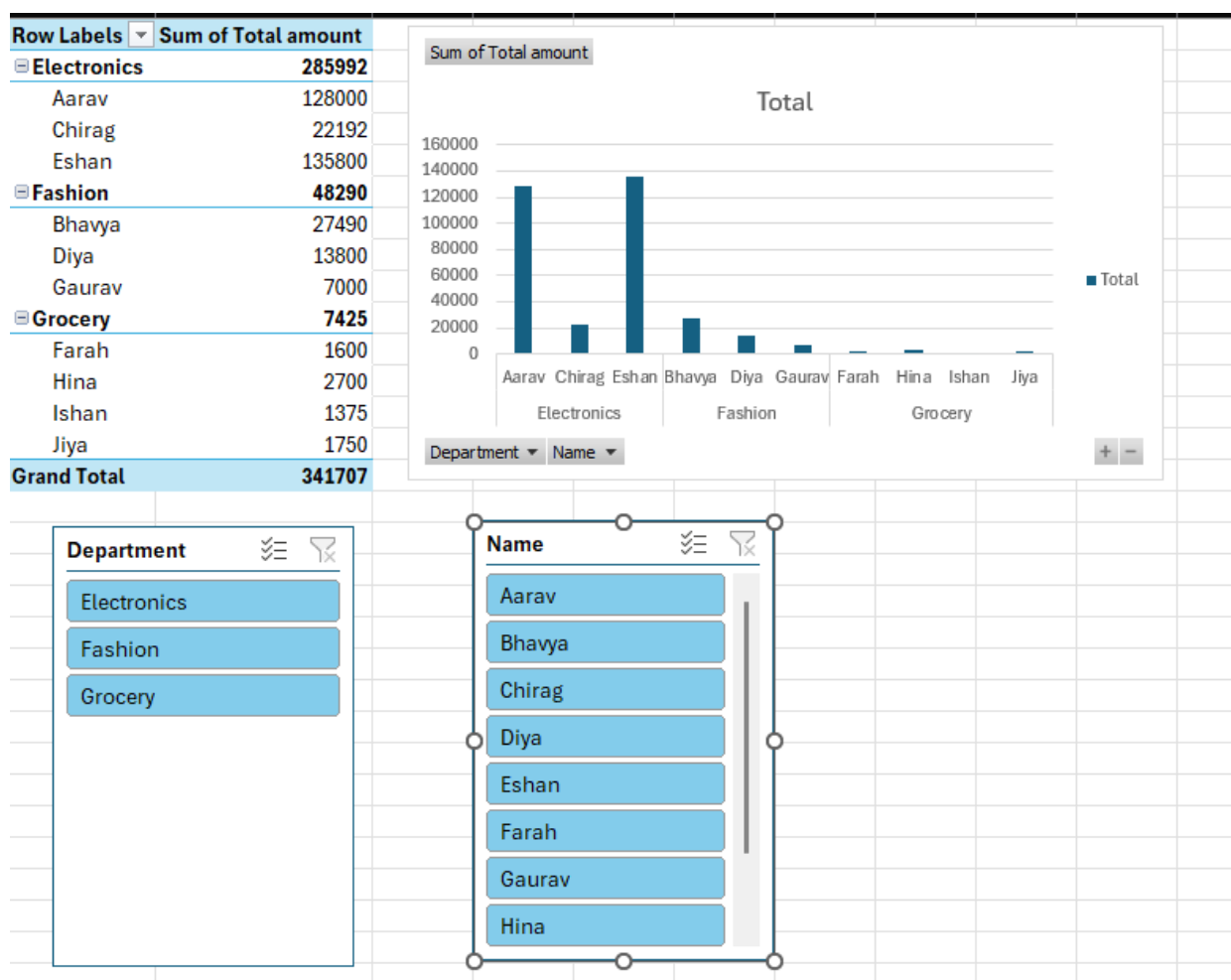
Task 2:

Screenshots of 3 Pivot Tables, 1 Pivot Chart, and the Slicer in use.

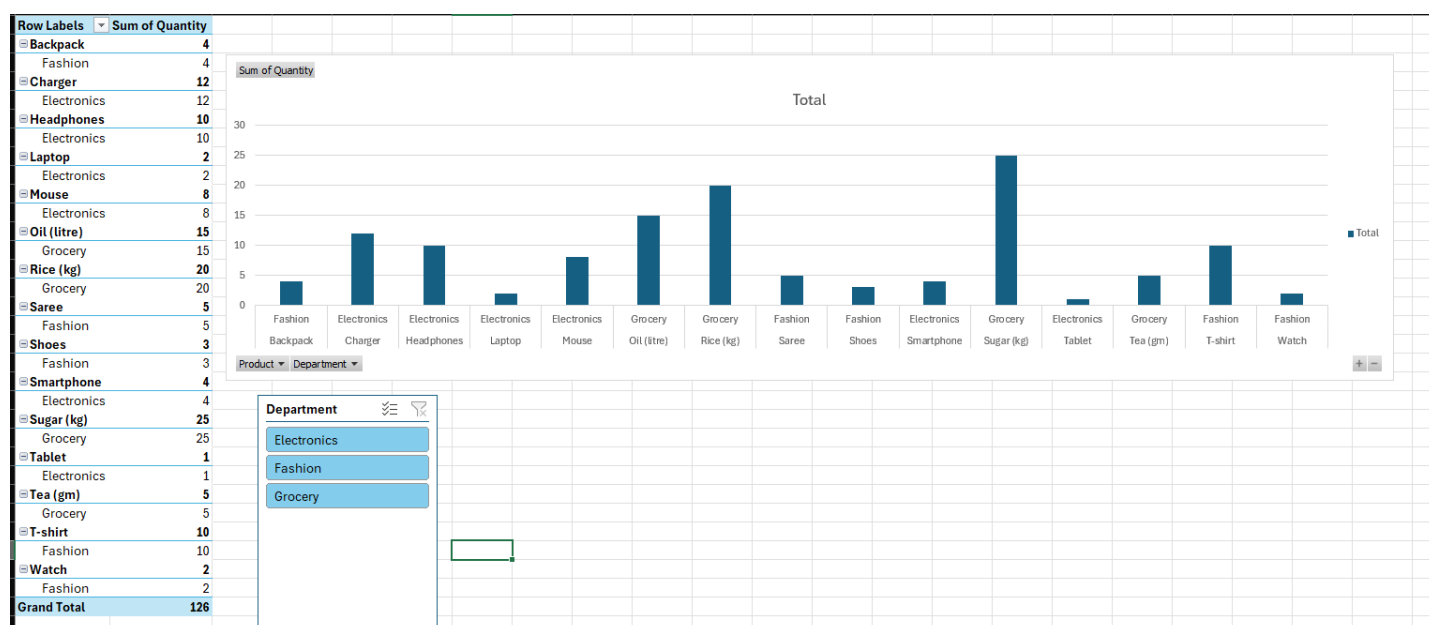
Pivot table 1:



Pivot table 2:



Pivot table 3:



Task 3:

All 10 SQL queries plus 3 CRUD statements with their outputs.

Sql query 1

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```
1 CREATE TABLE employees (  
2     emp_id INT PRIMARY KEY,  
3     name VARCHAR(50),  
4     department VARCHAR(30),  
5     salary INT,  
6     hire_date DATE  
7 );  
8  
9 INSERT INTO employees VALUES  
10 (1, 'Aarav', 'Electronics', 55000, '2022-01-10'),  
11 (2, 'Bhavya', 'Fashion', 48000, '2021-05-15'),  
12 (3, 'Chirag', 'Electronics', 52000, '2023-03-20'),  
13 (4, 'Diya', 'Fashion', 72000, '2020-07-01'),  
14 (5, 'Eshan', 'Electronics', 60000, '2022-11-12'),  
15 (6, 'Farah', 'Grocery', 45000, '2023-08-05'),  
16 (7, 'Gaurav', 'Fashion', 85000, '2019-04-22'),  
17 (8, 'Hina', 'Grocery', 42000, '2024-01-18'),  
18 (9, 'Ishan', 'Grocery', 55000, '2022-09-10'),  
19 (10, 'Jiya', 'Electronics', 78000, '2020-12-01');  
20  
21 CREATE TABLE sales (  
22     sale_id INT PRIMARY KEY,  
23     emp_id INT,  
24     product VARCHAR(50),  
25     amount INT,  
26     sale_date DATE  
27 );  
28  
29 INSERT INTO sales VALUES  
30 (101, 1, 'Laptop', 110000, '2025-02-03'),  
31 (102, 2, 'Saree', 22500, '2025-02-05'),  
32 (103, 3, 'Headphones', 15000, '2025-02-06'),  
33 (104, 4, 'Shoes', 6600, '2025-02-07'),  
34 (105, 5, 'Smartphone', 128000, '2025-02-09'),  
35 (106, 1, 'Tablet', 18000, '2025-02-11'),  
36 (107, 6, 'Rice', 1600, '2025-02-12'),  
37 (108, 7, 'Watch', 7000, '2025-02-14'),  
38 (109, 8, 'Oil', 2700, '2025-02-15'),  
39 (110, 3, 'Mouse', 7192, '2025-02-18');  
40  
41 SELECT * FROM employees;  
42 SELECT * FROM sales;
```

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STDIN

Input for the program (Optional)

Output:

emp_id	name	department	salary	hire_date
1	Aarav	Electronics	55000	2022-01-10
2	Bhavya	Fashion	48000	2021-05-15
3	Chirag	Electronics	52000	2023-03-20
4	Diya	Fashion	72000	2020-07-01
5	Eshan	Electronics	60000	2022-11-12
6	Farah	Grocery	45000	2023-08-05
7	Gaurav	Fashion	85000	2019-04-22
8	Hina	Grocery	42000	2024-01-18
9	Ishan	Grocery	55000	2022-09-10
10	Jiya	Electronics	78000	2020-12-01

sale_id	emp_id	product	amount	sale_date
101	1	Laptop	110000	2025-02-03
102	2	Saree	22500	2025-02-05
103	3	Headphones	15000	2025-02-06
104	4	Shoes	6600	2025-02-07
105	5	Smartphone	128000	2025-02-09
106	1	Tablet	18000	2025-02-11
107	6	Rice	1600	2025-02-12
108	7	Watch	7000	2025-02-14
109	8	Oil	2700	2025-02-15
110	3	Mouse	7192	2025-02-18

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Sql query 2

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```
38  
39 -- Query 1: Display all rows of employees  
40 SELECT * FROM employees;  
41  
42 -- Query 2: List employees from 'Fashion' department  
43 SELECT * FROM employees  
44 WHERE department = 'Fashion';  
45  
46 -- Query 3: Employees sorted by salary descending  
47 SELECT * FROM employees  
48 ORDER BY salary DESC;  
49  
50 -- Query 4: Employees with salary between 50000 and 70000  
51 SELECT name, salary FROM employees  
52 WHERE salary BETWEEN 50000 AND 70000;  
53  
54 -- Query 5: Count of employees in each department  
55 SELECT department, COUNT(*) AS employee_count  
56 FROM employees  
57 GROUP BY department;  
58  
59 -- Query 6: Average salary per department  
60 SELECT department, AVG(salary) AS avg_salary  
61 FROM employees  
62 GROUP BY department;  
63  
64 -- Query 7: Departments with average salary > 50000  
65 SELECT department, AVG(salary) AS avg_salary  
66 FROM employees  
67 GROUP BY department  
68 HAVING AVG(salary) > 50000;  
69  
70 -- Query 8: INNER JOIN to show emp_id, name, product, amount  
71 SELECT e.emp_id, e.name, s.product, s.amount  
72 FROM sales s  
73 INNER JOIN employees e ON s.emp_id = e.emp_id;  
74  
75 -- Query 9: Total sales amount made by each employee  
76 SELECT e.emp_id, e.name, SUM(s.amount) AS total_sales  
77 FROM employees e  
78 JOIN sales s ON e.emp_id = s.emp_id  
79 GROUP BY e.emp_id, e.name;  
80  
81 -- Query 10: Subquery - Employees with salary > average salary  
82 SELECT name, salary  
83 FROM employees  
84 WHERE salary > (SELECT AVG(salary) FROM employees);
```

I/O AI Agent

STDIN

Input for the program (Optional)

Output:

emp_id	name	department	salary	hire_date
1	Aarav	Electronics	55000	2022-01-10
2	Bhavya	Fashion	48000	2021-05-15
3	Chirag	Electronics	52000	2023-03-20
4	Diya	Fashion	72000	2020-07-01
5	Eshan	Electronics	60000	2022-11-12
6	Farah	Grocery	45000	2023-08-05
7	Gaurav	Fashion	85000	2019-04-22
8	Hina	Grocery	42000	2024-01-18
9	Ishan	Grocery	55000	2022-09-10
10	Jiya	Electronics	78000	2020-12-01

emp_id	name	department	salary	hire_date
2	Bhavya	Fashion	48000	2021-05-15
4	Diya	Fashion	72000	2020-07-01
7	Gaurav	Fashion	85000	2019-04-22

emp_id	name	department	salary	hire_date
7	Gaurav	Fashion	85000	2019-04-22
10	Jiya	Electronics	78000	2020-12-01
4	Diya	Fashion	72000	2020-07-01
5	Eshan	Electronics	60000	2022-11-12
1	Aarav	Electronics	55000	2022-01-10
9	Ishan	Grocery	55000	2022-09-10
3	Chirag	Electronics	52000	2023-03-20
2	Bhavya	Fashion	48000	2021-05-15
6	Farah	Grocery	45000	2023-08-05
8	Hina	Grocery	42000	2024-01-18

name	salary
Aarav	55000
Chirag	52000
Eshan	60000
Ishan	55000

department	employee_count
Electronics	5
Fashion	3
Grocery	3

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Output:

department	employee_count
------------	----------------

Electronics	4
Fashion	3
Grocery	3

department	avg_salary
------------	------------

Electronics	61250
Fashion	68333
Grocery	47333

department	avg_salary
------------	------------

Electronics	61250
Fashion	68333

emp_id	name	product	amount
1	Aarav	Laptop	110000
2	Bhavya	Saree	22500
3	Chirag	Headphones	15000
4	Diya	Shoes	6600
5	Eshan	Smartphone	128000
1	Aarav	Tablet	18000
6	Farah	Rice	1600
7	Gaurav	Watch	7000
8	Hina	Oil	2700
3	Chirag	Mouse	7192

emp_id	name	total_sales
1	Aarav	128000
2	Bhavya	22500
3	Chirag	22192
4	Diya	6600
5	Eshan	128000
6	Farah	1600
7	Gaurav	7000
8	Hina	2700

emp_id	name	total_sales
1	Aarav	128000
2	Bhavya	22500
3	Chirag	22192
4	Diya	6600
5	Eshan	128000
6	Farah	1600
7	Gaurav	7000
8	Hina	2700

name	salary
Diya	72000
Eshan	60000
Gaurav	85000
Jiya	78000

Sql query 3

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```
15 (7, 'Gaurav', 'Fashion', 85000, '2019-04-22'),
16 (8, 'Hina', 'Grocery', 42000, '2024-01-18'),
17 (9, 'Ishan', 'Grocery', 55000, '2022-09-10'),
18 (10, 'Jiya', 'Electronics', 78000, '2020-12-01');
19
20 CREATE TABLE sales (
21     sale_id INT PRIMARY KEY,
22     emp_id INT,
23     product VARCHAR(50),
24     amount INT,
25     sale_date DATE
26 );
27 INSERT INTO sales VALUES
28 (101, 1, 'Laptop', 110000, '2025-02-03'),
29 (102, 2, 'Saree', 22500, '2025-02-05'),
30 (103, 3, 'Headphones', 15000, '2025-02-06'),
31 (104, 4, 'Shoes', 6600, '2025-02-07'),
32 (105, 5, 'Smartphone', 128000, '2025-02-09'),
33 (106, 1, 'Tablet', 18000, '2025-02-11'),
34 (107, 6, 'Rice', 1600, '2025-02-12'),
35 (108, 7, 'Watch', 7000, '2025-02-14'),
36 (109, 8, 'Oil', 2700, '2025-02-15'),
37 (110, 3, 'Mouse', 7192, '2025-02-18');
38
39 SELECT * FROM employees;
40
41 -- CRUD 1: INSERT new employee
42 INSERT INTO employees (emp_id, name, department, salary, hire_date)
43 VALUES (11, 'Kavya', 'Fashion', 50000, '2025-03-01');
44
45 SELECT * FROM employees WHERE emp_id = 11;
46
47 -- CRUD 2: UPDATE salary of 'Grocery' department by 10%
48 UPDATE employees
49 SET salary = salary * 1.10
50 WHERE department = 'Grocery';
51
52 SELECT * FROM employees WHERE department = 'Grocery';
53
54 -- CRUD 3: DELETE employee with emp_id = 11
55 DELETE FROM employees
56 WHERE emp_id = 11;
57
58 SELECT * FROM employees WHERE emp_id = 11;
```

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I/O AI Agent

STDIN

Input for the program (Optional)

Output:

emp_id	name	department	salary	hire_date
1	Aarav	Electronics	55000	2022-01-10
2	Bhavya	Fashion	48000	2021-05-15
3	Chirag	Electronics	52000	2023-03-20
4	Diya	Fashion	72000	2020-07-01
5	Eshan	Electronics	60000	2022-11-12
6	Farah	Grocery	45000	2023-08-05
7	Gaurav	Fashion	85000	2019-04-22
8	Hina	Grocery	42000	2024-01-18
9	Ishan	Grocery	55000	2022-09-10
10	Jiya	Electronics	78000	2020-12-01

emp_id	name	department	salary	hire_date
11	Kavya	Fashion	50000	2025-03-01

emp_id	name	department	salary	hire_date
6	Farah	Grocery	49500	2023-08-05
8	Hina	Grocery	46200	2024-01-18
9	Ishan	Grocery	60500	2022-09-10

emp_id	name	department	salary	hire_date
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SQL Queries:

-- Database Setup

CREATE TABLE employees (

emp_id INT PRIMARY KEY,

name VARCHAR(50),

department VARCHAR(30),

salary INT,

hire_date DATE

);

INSERT INTO employees VALUES

(1, 'Aarav', 'Electronics', 55000, '2022-01-10'),

(2, 'Bhavya', 'Fashion', 48000, '2021-05-15'),

(3, 'Chirag', 'Electronics', 52000, '2023-03-20'),

(4, 'Diya', 'Fashion', 72000, '2020-07-01'),

(5, 'Eshan', 'Electronics', 60000, '2022-11-12'),

(6, 'Farah', 'Grocery', 45000, '2023-08-05'),

(7, 'Gaurav', 'Fashion', 85000, '2019-04-22'),

(8, 'Hina', 'Grocery', 42000, '2024-01-18'),

(9, 'Ishan', 'Grocery', 55000, '2022-09-10'),

(10, 'Jiya', 'Electronics', 78000, '2020-12-01');

CREATE TABLE sales (

sale_id INT PRIMARY KEY,

emp_id INT,

```
product VARCHAR(50),  
amount INT,  
sale_date DATE  
);  
  
INSERT INTO sales VALUES  
(101, 1, 'Laptop', 110000, '2025-02-03'),  
(102, 2, 'Saree', 22500, '2025-02-05'),  
(103, 3, 'Headphones', 15000, '2025-02-06'),  
(104, 4, 'Shoes', 6600, '2025-02-07'),  
(105, 5, 'Smartphone', 128000, '2025-02-09'),  
(106, 1, 'Tablet', 18000, '2025-02-11'),  
(107, 6, 'Rice', 1600, '2025-02-12'),  
(108, 7, 'Watch', 7000, '2025-02-14'),  
(109, 8, 'Oil', 2700, '2025-02-15'),  
(110, 3, 'Mouse', 7192, '2025-02-18');
```

10 Sql Queries:

-- Query 1: Display all rows of employees

```
SELECT * FROM employees;
```

-- Query 2: List employees from 'Fashion' department

```
SELECT * FROM employees
```

```
WHERE department = 'Fashion';
```

-- Query 3: Employees sorted by salary descending

```
SELECT * FROM employees
```

```
ORDER BY salary DESC;
```

-- Query 4: Employees with salary between 50000 and 70000

```
SELECT name, salary FROM employees
```

```
WHERE salary BETWEEN 50000 AND 70000;
```

-- Query 5: Count of employees in each department

```
SELECT department, COUNT(*) AS employee_count
```

```
FROM employees
```

```
GROUP BY department;
```

-- Query 6: Average salary per department

```
SELECT department, AVG(salary) AS avg_salary  
FROM employees  
GROUP BY department;
```

-- Query 7: Departments with average salary > 50000

```
SELECT department, AVG(salary) AS avg_salary  
FROM employees  
GROUP BY department  
HAVING AVG(salary) > 50000;
```

-- Query 8: INNER JOIN to show emp_id, name, product, amount

```
SELECT e.emp_id, e.name, s.product, s.amount  
FROM sales s  
INNER JOIN employees e ON s.emp_id = e.emp_id;
```

-- Query 9: Total sales amount made by each employee

```
SELECT e.emp_id, e.name, SUM(s.amount) AS total_sales  
FROM employees e  
JOIN sales s ON e.emp_id = s.emp_id  
GROUP BY e.emp_id, e.name;
```

```
-- Query 10: Subquery - Employees with salary > average salary  
SELECT name, salary  
FROM employees  
WHERE salary > (SELECT AVG(salary) FROM employees);
```

3 CRUD Operations:

-- CRUD 1: INSERT new employee

```
INSERT INTO employees (emp_id, name, department, salary, hire_date)
VALUES (11, 'Kavya', 'Fashion', 50000, '2025-03-01');
```

```
SELECT * FROM employees WHERE emp_id = 11;
```

-- CRUD 2: UPDATE salary of 'Grocery' department by 10%

```
UPDATE employees
```

```
SET salary = salary * 1.10
```

```
WHERE department = 'Grocery';
```

```
SELECT * FROM employees WHERE department = 'Grocery';
```

-- CRUD 3: DELETE employee with emp_id = 11

```
DELETE FROM employees
```

```
WHERE emp_id = 11;
```

```
SELECT * FROM employees WHERE emp_id = 11;
```

SQL Queries (Task 3.1):

#	Task / Query Description	SQL Query
1	Display all rows of the employees table	SELECT * FROM employees;
2	List employees from the 'Fashion' department only	SELECT * FROM employees WHERE department = 'Fashion';
3	Display all employees sorted by salary in descending order	SELECT * FROM employees ORDER BY salary DESC;
4	Display names of employees with salary between 50000 and 70000	SELECT name, salary FROM employees WHERE salary BETWEEN 50000 AND 70000;
5	Find the total number of employees in each department	SELECT department, COUNT(*) AS employee_count FROM employees GROUP BY department;
6	Find the average salary per department	SELECT department, AVG(salary) AS avg_salary FROM employees GROUP BY department;
7	Find departments where average salary is greater than 50000	SELECT department, AVG(salary) AS avg_salary FROM employees GROUP BY department HAVING AVG(salary) > 50000;
8	INNER JOIN to show emp_id, name, product, and amount for all sales	SELECT e.emp_id, e.name, s.product, s.amount FROM sales s INNER JOIN employees e ON s.emp_id = e.emp_id;
9	Find total sales amount made by each employee	SELECT e.emp_id, e.name, SUM(s.amount) AS total_sales FROM employees e JOIN sales s ON e.emp_id = s.emp_id GROUP BY e.emp_id, e.name;

10	Subquery to find employees with salary greater than average salary	SELECT name, salary FROM employees WHERE salary > (SELECT AVG(salary) FROM employees);
-----------	--	--

CRUD Operations (Task 3.2):

#	Operation	SQL Statement	Verification Query
1	INSERT: Add employee (emp_id=11, Kavya, Fashion, 50000, 2025-03-01)	INSERT INTO employees (emp_id, name, department, salary, hire_date) VALUES (11, 'Kavya', 'Fashion', 50000, '2025-03-01');	SELECT * FROM employees WHERE emp_id = 11;
2	UPDATE: Increase salary of 'Grocery' department by 10%	UPDATE employees SET salary = salary * 1.10 WHERE department = 'Grocery';	SELECT * FROM employees WHERE department = 'Grocery';
3	DELETE: Remove employee with emp_id=11	DELETE FROM employees WHERE emp_id = 11;	SELECT * FROM employees WHERE emp_id = 11;